

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Python** data pipelines on **AWS Lambda** and **S3** for large-scale AI model pre-training, processing ~3,500 events/sec during peak hours and replacing a legacy batch system.
- Developed and optimized **distributed training pipelines** for transformer architectures using **PyTorch** across large GPU infrastructures, focusing on scalability and computational efficiency.
- Researched and prototyped novel **model architectures** for multi-modal AI systems, conducting independent experiments and analyzing results to iterate on methodologies for improved quality.
- Containerized **AI training services** with **Docker** and deployed them onto an **AWS EKS** cluster, configuring **Helm charts** and resource limits for predictable rollouts and operational stability.
- Contributed to data curation strategies and baseline improvements for large-scale pre-training datasets, enhancing model training outcomes and strengthening generalization capabilities.
- Identified and resolved computational bottlenecks in **model performance** and scalability through deep dives into **distributed machine learning systems** and GPU utilization analysis.
- Collaborated with **cross-functional engineering and research teams** to accelerate innovation, integrating new model architectures and training methodologies into existing systems.

SKILLS

AI/ML Frameworks – PyTorch, TensorFlow, JAX, scikit-learn

Programming Languages – Python, Java, SQL, R

Cloud & Distributed Systems – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, EKS, distributed training, GPU infrastructure

AI Concepts – Large Language Models (LLMs), Transformer Architectures, AI Pre-training, Multi-modal AI, Deep Learning Optimization, Model Scaling, Performance Tuning, Data Curation

Development Tools – Git, Jenkins, Jira, Confluence, Helm

Data & Databases – SQL (PostgreSQL, MySQL, Oracle), pandas, NumPy

PROJECTS

Large-Scale Transformer Pre-training Pipeline (HCL Technologies)

- Engineered a distributed training pipeline in **Python** on **AWS** to pre-train transformer-based language models, handling datasets exceeding 500GB.
- Implemented data parallelism and model parallelism strategies within **PyTorch** to optimize training across a cluster of **GPUs**, reducing average training time by 18%.
- Developed monitoring scripts using **CloudWatch** to track training progress, resource utilization, and identify performance bottlenecks in near real-time.

AI Model Inference Service Deployment (HCL Technologies)

- Built a **FastAPI** service in **Python** to serve inference requests for a sentiment analysis model, achieving p95 latency under 150ms for ~500 requests per minute.
- Containerized the inference service using **Docker** and deployed it to a **Kubernetes** cluster, enabling automated scaling based on request volume.
- Integrated request validation and error handling mechanisms to ensure reliable model predictions in a production-like environment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)